README: Build your first agent in Microsoft Foundry

TechSnacks 1 · Guided Hands-on Lab · new Microsoft Foundry portal

This guide walks you through the lab one click at a time. You do not need any coding or previous Foundry experience. Follow the steps in order, tick each box as you go, and raise your hand (or write in the chat) whenever you get stuck. Rajat and Derya from Microsoft are there to help.

New portal: *this guide follows the **new** Microsoft Foundry experience. At the top of https://ai.azure.com, make sure the **New Foundry** toggle is **on**. If your screen shows a "Setup" pane on the right with a "Knowledge" section, you are in the classic portal. Switch the toggle on.*

Labels may vary: *Foundry is updated often. If a button has a slightly different name, choose the closest match. The steps stay the same.*

What you will build

You will build an AI agent in Microsoft Foundry, give it instructions, ground it in a document, test it, look at what happened behind the scenes, and (if you have the right permissions) publish it.

You can choose one of two paths:

	Path A: Sample scenario (recommended)	**Path B: Bring your own idea**
What	"Cloud Guide", a helpdesk agent for developers at the fictional company Contoso Energy	A small agent for a use case in your own team
Knowledge	The sample document "Contoso Energy – Azure Cloud Usage Guidelines" (Appendix B)	1–3 of your own documents, non-confidential only
Instructions	Ready to paste (Appendix A)	Fill in the template (Appendix C)
Tool (optional)	Microsoft Learn MCP server (free)	Optional

If you are unsure, take **Path A**. Everything you need is in this guide.

Words you will see

Term	**What it means in plain words**
Microsoft Foundry	Microsoft's web portal for building AI apps and agents, at https://ai.azure.com
Project	Your workspace in Foundry. It holds your

Term	What it means in plain words
	models, agents and files
Build	The menu at the top right of the portal where you find Models and Agents
Model	The "brain" of the agent, for example gpt-5-mini
Deployment	A copy of a model switched on for your project, so your agent can use it
Agent	The model plus your instructions, knowledge and tools, working towards a goal
Playground	The agent page: settings on the left, a chat on the right to test your agent
Instructions	The text that tells the agent who it is, what to do and what not to do
File search	The tool that lets the agent search your uploaded documents (grounding)
Tool	Something the agent can use to act or look things up, for example web search or an MCP server
MCP	Model Context Protocol: an open standard for connecting agents to tools
Version	Every time you save the agent, Foundry saves a new version
Traces	A record of what the agent did behind the scenes to produce an answer

Step 0: Before you start

Check that you have the following. Without them you can follow along, but you will not be able to build.

- ☐ Access to a DEV or TEST Azure subscription (request it via the IT Service Portal at least one week before)
- ☐ **Contributor** access on the Azure resource group you will use
- ☐ The **Foundry User** role on the Foundry project (formerly called "Azure AI User")
- ☐ To publish in Step 8: permission to create an Azure Bot Service resource in the resource group (Contributor covers this). Your organisation may also require the **Foundry Project Manager** role
- ☐ You can open https://ai.azure.com in your browser and sign in with your company account
- ☐ For Path A: the sample document saved as a file on your computer (see Appendix B)

Costs and data: *Please check with your subscription owner before using a subscription. Everything you create costs a little money and is the responsibility of the subscription owner, so follow the clean-up step at the end. Use only sample or non-confidential data in this lab.*

Support: *For technical support, contact Chris Smith (Infrastructure) or the Azure Platform team.*

Step 1: Create a project

1. Open https://ai.azure.com and sign in with your company account.
2. Make sure the **New Foundry** toggle at the top of the page is **on**.
3. Open the project selector in the upper-left corner and select **Create new project**. (If this is your first time in Foundry, select **Create a new project** on the welcome page.)
4. Enter a **Project name**, for example `techsnack-lab-yourname` (lowercase, no spaces).
5. Select **Advanced options** and set the values in the table below.
6. Select **Create** and wait. If a welcome dialog appears, select **Let's go**.

Setting	Value
Subscription	Your DEV or TEST subscription
Resource group	Your resource group
Region	A region allowed by your Azure Platform team, for example Sweden Central

- ☐ **Done when:** you see your project's home page with its name in the upper-left corner.

Step 2: Deploy a model

1. Select **Build** (top right), then **Models** in the left menu.
2. Select **Deploy** > **Deploy a base model**.
3. Search for `gpt-5-mini` and select it.
4. Select **Deploy** > **Default settings** (or **Custom settings** if your Azure Platform team asks for a specific deployment type, for example **Data Zone Standard** to keep processing in the EU).
5. Wait until the deployment is ready.

If you see a quota or region error, try `gpt-4.1-mini` instead, or ask us for help. Small "mini" models are fast, cheap and good enough for most agent tasks.

- ☐ **Done when:** the model appears in your list of deployments.

Step 3: Create your agent

1. Select **Build** > **Agents**, then **Create agent** (or, on the project home page, under **Build an agent**, select **Create an agent**).
2. Enter the **Agent name**: `cloud-guide` (Path A) or your own name (Path B).
3. Keep **Interaction mode** set to **Text**.
4. Leave **Use the Microsoft Copilot harness** unselected.
5. Select **Create**. The agent playground opens: settings on the left, chat on the right.
6. In **Model**, select the deployment from Step 2.

7. **Remove web search:** look at the **Tools** section. If **Web search** is listed, open its ⋮ menu and select **Remove**. Web search sends data to Bing, outside the Azure compliance boundary, and would mix public web answers into your test.

- ☐ **Done when:** your model is selected and the Tools section has no web search.

Step 4: Write the instructions

Instructions are the most important part of your agent. They tell it its role, which knowledge to use, its boundaries and how to answer.

1. **Path A:** copy the full text from Appendix A and paste it into the **Instructions** box.
2. **Path B:** fill in the template from Appendix C and paste it into the **Instructions** box.
3. Select **Save**. Foundry saves this as a new version of your agent.

If you will not do the optional tool step (Step 7), delete the Microsoft Learn bullet from the Path A instructions. Otherwise the agent may offer to look things up but cannot.

- ☐ **Done when:** your instructions are saved.

Step 5: Add knowledge (file search)

1. In the **Tools** section of the playground, select **Add** > **Add tools**.
2. Select **File search**.
3. Upload your document. For Path A this is the sample document from Appendix B. For Path B, upload 1–3 non-confidential documents (for example .docx, .pdf, .md or .txt).
4. Foundry creates a vector store (a searchable index of your file). Give it a name if asked, for example `cloud-guidelines`, and select **Add** (or **Upload and save**).
5. Wait until the file shows as ready, then select **Save** (or **Save and start new chat**).

- ☐ **Done when: File search** is listed under Tools with your file attached.

Step 6: Test your agent

1. Use the chat on the right side of the playground. If you just saved, select **Save and start new chat** (or **New chat**) so the agent uses the latest version.
2. Ask the test questions below one at a time and compare the answers with the expected behaviour.
3. Check that answers from the document show a citation (a reference to the file).

Ask this (Path A)	**A good answer…**
Which regions can I use for production?	Germany West Central (primary) and West Europe (secondary/DR), quoting section 2
What should I name a storage account for the WindOps app in DEV in Germany West Central?	`cewindopsdevgwcst`, explaining how the name was built
Can I put customer data in my DEV environment?	No: DEV/TEST must not contain confidential or personal data; use synthetic data
What's on the canteen menu today?	Politely says it cannot help with that

Path B: write three questions of your own. Two should be answerable from your documents, and one should be something the agent must refuse.

*If the agent answers from general knowledge instead of the document, check that **File search** is listed under Tools, that the file is ready, that you saved and started a new chat, and that your instructions say to answer only from the document.*

- ☐ **Done when:** one grounded answer with a citation, and one polite refusal.

Step 7 (optional stretch): Add a free MCP tool

This lets your agent look up official Microsoft documentation. It is free and needs no sign-in or key.

1. In the **Tools** section, select **Add** > **Add tools**.
2. Open the **Custom** tab and select **Model Context Protocol (MCP)**.
3. Enter the details from the table below and select **Connect** (or **Add**).
4. Select **Save and start new chat**.
5. Ask: *"How do I enable soft delete for blobs on an Azure storage account? Check the official docs."*
6. Foundry may ask you to **approve** the tool call. Look at what the agent wants to send, then approve. This is "human in the loop": you stay in control of what the agent does.
7. A good answer gives the steps and a link to Microsoft Learn.

Field	Value
Name	`mslearn`
Remote MCP server endpoint	`https://learn.microsoft.com/api/mcp`
Authentication	None (unauthenticated)

Bonus question that uses both knowledge and the tool: *"I want to deploy gpt-5-mini in Sweden Central for a DEV test. Is that allowed, and how do I do it?"*

- ☐ **Done when:** the answer includes a Microsoft Learn link.

Step 8: Look at the traces and publish

Traces

1. Under one of the agent's answers in the chat, select **Traces**.
2. You might have to enable App Insight if not already done.
    a. Create or connect an App Insights resource to enable tracing.
3. After App Insights is enabled Ask your agent similar question *"I want to deploy gpt-5-mini in Sweden Central for a DEV test. Is that allowed, and how do I do it?"*
4. Look at the steps: which tools the agent called (for example **file search** or **mslearn**), how long it took and how many tokens it used. This is how you troubleshoot an agent.

You can see these response details without any extra setup. Full end-to-end tracing and monitoring need Application Insights connected to the project. That's optional for this lab.

Publish (optional)

1. At the top of the agent page, select **Publish**.
2. Next to **Active version**, keep **Always use latest** (or pick the version you tested).
3. Choose **Publish to Microsoft 365 and Teams** and fill in the name and short description. For a test, placeholder website and privacy URLs are fine.
4. Select **Prepare agent** and wait 1–2 minutes, then follow the prompts to make it available in Teams and Microsoft 365 Copilot.

If publishing fails or the button is greyed out, you are missing a permission (for example to create an Azure Bot Service resource). That's fine: you have still built and tested a working agent.

- ☐ **Done when:** you have opened at least one trace.

Step 9: Clean up (important)

Resources cost money while they exist. When you have finished experimenting, do one of the following:

- In Foundry, delete your agent (**Build** > **Agents**), your model deployment (**Build** > **Models**) and, if nobody else uses it, the project.
- Or, if you created a resource group only for this lab, delete the whole resource group in the Azure portal (https://portal.azure.com). Only do this if nothing else is in it.

Troubleshooting

Problem	What to try
My screen looks different from this guide	Check that the **New Foundry** toggle is on. If you see a "Setup" pane on the right, you are in the classic portal
I can't see my subscription or resource group	Check you're in the right directory (top-right profile). Check your roles with the Azure Platform team
Quota or "model not available in region" error	Try another region or `gpt-4.1-mini`
The deployment takes a long time	Wait a moment and refresh the page
The agent answers from the web	Remove **Web search** from the Tools section, then **Save and start new chat**
The agent ignores my document	Check **File search** is listed under Tools, the file is ready, you saved, and the instructions say to answer only from the document
The agent makes things up	Add "Never invent facts. If it's not in the document, say so." to the instructions
Changes don't seem to work	Select **Save and start new chat**. Old chats keep using the previous version
The MCP tool isn't used	Ask explicitly ("check the official docs"), and

Problem	What to try
	approve the tool call when asked
Publishing fails	You need permission to create an Azure Bot Service resource. Ask the Azure Platform team
Labels don't match this guide	The portal changes often. Look for the closest match or ask us

What's next

Session	Date	Topic
2: Engineer an Agent	19 Nov 2026, 15:00–17:00	From Foundry to Agent Framework: code-first agents, tools, memory and multi-agent
3: Ground an Agent	3 Dec 2026, 15:00–17:00	Grounding with enterprise knowledge: RAG, Azure AI Search, Document Intelligence
4: Connect an Agent	17 Dec 2026, 15:00–17:00	Connecting agents to tools and APIs with MCP

Useful links

- Microsoft Foundry portal: https://ai.azure.com
- Foundry documentation: https://learn.microsoft.com/azure/foundry/
- Agents overview: https://learn.microsoft.com/azure/foundry/agents/overview
- File search tool: https://learn.microsoft.com/azure/foundry/agents/how-to/tools/file-search

Appendix A: Sample agent instructions (Path A)

Copy everything in the box and paste it into the **Instructions** box.

```
# Role
You are "Cloud Guide", a friendly cloud helpdesk assistant for software
developers at Contoso Energy. You help them use Azure correctly and in
line with company rules.

# Knowledge and tools
- For questions about internal rules (regions, naming, tags, costs,
data, AI models, security, access requests), answer ONLY from the
uploaded document "Contoso Energy – Azure Cloud Usage Guidelines".
Always mention the section you used.
- For technical how-to questions about Azure or Microsoft products, use
the Microsoft Learn tool (mslearn) and include the link to the page you
used.
- If a question needs both, first check the internal rule, then explain
how to do it using Microsoft Learn.

# Boundaries
```

```
- Never invent policies, regions, names, prices or contacts. If the
guidelines don't cover it, say so and suggest contacting the Cloud
Platform team (cloud-platform@contoso-energy.example).
- If a question is not about Azure or Contoso Energy's cloud rules,
politely say you can't help with that.
- Never ask for or process confidential or personal data.

# Naming requests
- When asked to name a resource, apply the naming convention from the
guidelines and briefly show how you built the name.

# Style
- Keep answers short and practical: a one-line answer first, then up to
5 bullet points.
- Answer in English; if the user writes in German, answer in German.
```

Appendix B: Sample grounding document (Path A)

If you have not received the file "Contoso Energy – Azure Cloud Usage Guidelines (sample).docx", copy the text from the box below into a new Word document (or a .txt/.md file) and save it under that name. Then upload it in Step 5.

```
Contoso Energy – Azure Cloud Usage Guidelines
Version 1.0 · Owner: Cloud Platform Team
FICTIONAL DOCUMENT FOR TRAINING PURPOSES ONLY

1 Purpose and scope
These guidelines apply to all Azure workloads of Contoso Energy IT.

2 Approved regions
Production: Germany West Central (primary), West Europe (secondary /
disaster recovery).
DEV and TEST: Germany West Central, West Europe, Sweden Central.
Any other region requires approval from the Cloud Platform team.

3 Naming convention
Pattern: <org>-<app>-<env>-<regionCode>-<resourceType>
org = ce. env = dev, tst or prd.
Region codes: gwc (Germany West Central), weu (West Europe), sdc
(Sweden Central).
Resource type codes: rg (resource group), vnet (virtual network), kv
(Key Vault), app (App Service), func (Function App), aif (AI Foundry
resource), st (storage account).
Storage accounts: no hyphens, lowercase only, maximum 24 characters.
Examples: ce-windops-dev-gwc-rg, ce-windops-prd-gwc-kv, ce-gridmon-tst-
weu-func, cewindopsdevgwcst.

4 Mandatory tags
```

```
Every resource must have these tags: CostCenter, Owner (email address),
Environment, DataClassification (Public, Internal or Confidential),
Application.
Untagged resources are reported weekly and may be deleted in DEV after
14 days.

5 Environments and data
DEV and TEST must never contain confidential or personal data. Use
synthetic data.
Changes to PROD are made only through the deployment pipeline.

6 Cost rules
Budget alert: DEV 500 EUR per month per subscription, TEST 1,000 EUR
per month.
DEV virtual machines and compute shut down automatically at 19:00 CET.
Delete unused resources within 30 days.

7 AI services
The default model is gpt-5-mini.
Larger models (for example gpt-5 or reasoning models) need approval
from the AI Center of Excellence.
Do not put confidential data in prompts in DEV. Content filters must
stay enabled.

8 Security
Store secrets only in Azure Key Vault. Prefer managed identities.
Public network access is disabled for PROD storage accounts.

9 Requests and support
Request new subscriptions or role assignments with the IT Service
Portal ticket "Cloud Access Request" (SLA: 2 business days).
For urgent issues, contact the Cloud Platform on-call via the service
desk.
Contact: cloud-platform@contoso-energy.example

10 FAQ
Can I use East US for a DEV test? No, approval from the Cloud Platform
team is required.
Who pays for my DEV resources? The cost center in the CostCenter tag.
Can I use gpt-5 in DEV? Only with approval from the AI Center of
Excellence.
How long do I wait for access? 2 business days.
```

Appendix C: Instruction template (Path B)

Replace everything in [square brackets] with your own text, then paste it into the **Instructions** box.

```
# Role
```

```
You are [agent name], a [tone, e.g. friendly / precise] assistant that
helps [audience] with [task].

# Knowledge
Answer questions about [topic] only from the uploaded [document
name(s)] and mention the section you used.

# Tools (optional)
Use [tool] when [situation].

# Boundaries
If the answer is not in the documents, say so and point to
[contact/team]. Don't answer questions about [out-of-scope topics].
Never invent facts.

# Format
[e.g. short answer first, then bullet points; language].
```

Example: *You are "Shift Buddy", a precise assistant that helps wind-farm technicians find maintenance steps in the uploaded sample maintenance guideline.*

Good instructions checklist

- ☐ Clear role and audience
- ☐ Says which knowledge to use and when to use tools
- ☐ Says what to do when the answer isn't known
- ☐ Defines what's out of scope
- ☐ Defines answer format and language